LAB 4 — CONNECTIVITY OVERVIEW

Front and rear are the terminated ends. Cockpit taps both backbones.
Each zone has its own two isolation groups; termination remains on the bus side.
Actuator connectors / motor protection / physical layout remain unfinished.

Front zone



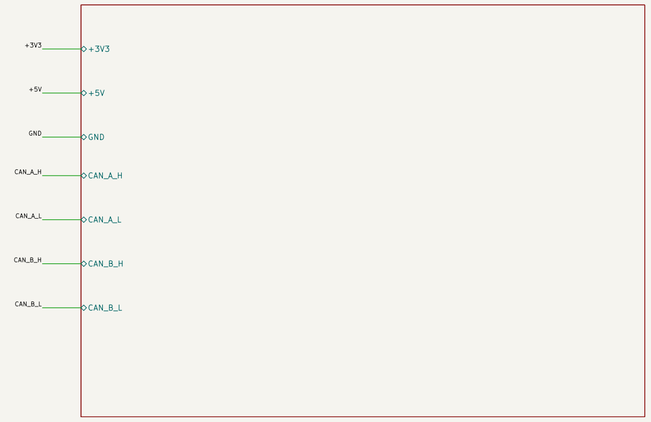
File: Front.kicad_sch

Cockpit zone



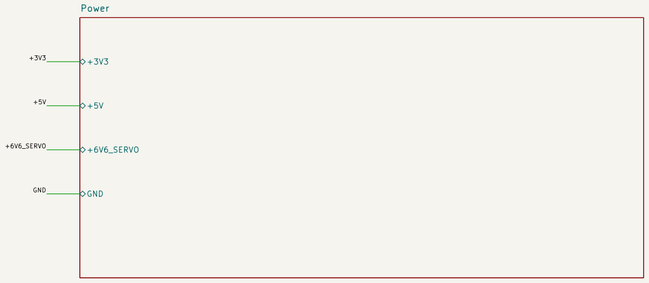
File: Cockpit.kicad_sch

Rear zone



File: Rear.kicad_sch

Power



File: Power.kicad_sch

CAN_A_H/L and CAN_B_H/L are separate root nets. Same-name local labels on this root join the shown sheet pins. Child labels connect only through those ports.

Power flags document external battery power and outputs after passive filters; they do not prove current capacity, regulation, thermal behavior or PCB completeness.

See Verification.txt and Design-review.csv for current checks and open decisions.

Shared nets: two CAN backbones, power and ground only.
PRELIMINARY — incomplete board; not for fabrication

Team 4

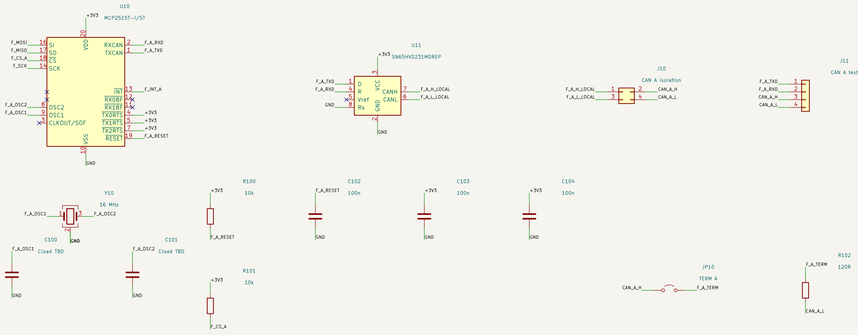
Sheet: /
File: Lab4.kicad_sch

Title: Lab 4 — three-zone controller

Size: A2	Date: 2026-10-20	**Rev: 0.3**
KiCad E.D.A. 10.0.4		Id: 1/5

Front zone — CAN and servo current sensing

CAN A: 16 MHz / 500 kbit/s / 3.3 V



RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

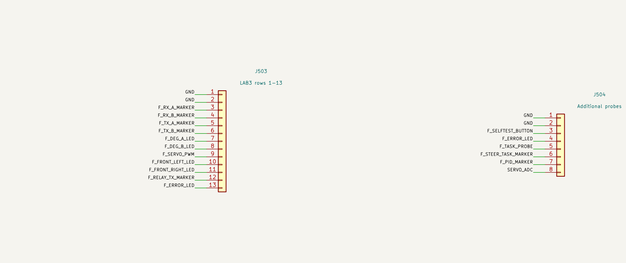
CAN B: 16 MHz / 500 kbit/s / 3.3 V



RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

SERVO SHUNT — calculated proposal (not measured)

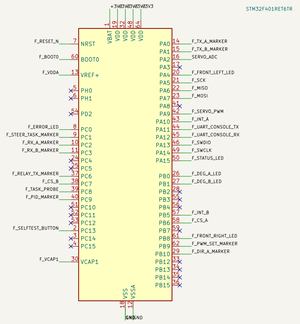
6.6 V regulator -> 10 mOhm shunt -> servo +
Servo GND returns directly to power ground.
Kelvin sense: separate thin traces from shunt pads to IN+/IN−.
Gain 50: Vout = 0.5 V/A. Proposed range 0–5 A -> 0–2.5 V.
12-bit / 3.3 V: 1.61 mA/count (ideal quantization).
At 5 A: shunt drop 50 mV; loss 0.25 W. Select >=1 W.
RC: 4.7k + 100n -> 339 Hz; proposed ADC 4 ksample/s.
Near-zero amplifier output clips: calibrate offset and gain.
This unidirectional circuit cannot measure regenerated current.
No servo Idle/stall/noise measurements yet.
5 A measurement range is NOT a 5 A driver or regulator rating.



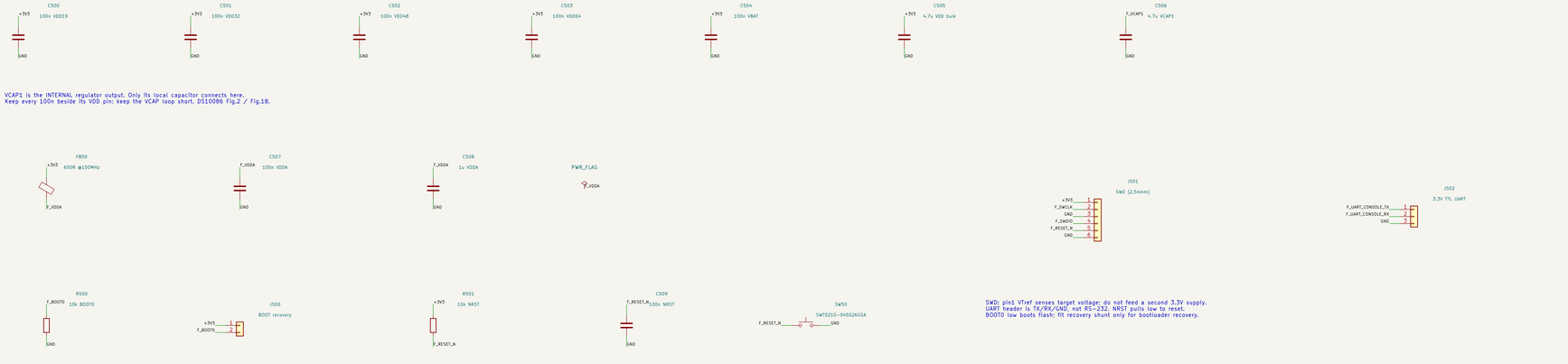
LAB3 header follows handout p17 top-to-bottom. Two GND pins first.
SRV/PWM_OUT probe actual actuator PWM; timing markers remain separate.
Physical header position / accessibility will be checked during PCB layout.

Servo connector not assigned yet; verify plug order/contact rating.
Power is SERVO_PLUS after the shunt; signal F_SERVO_PWM; return GND.
Front lamps: PA4 / PB8, output drivers and connectors still pending.

STM32F401RET6 — independent HSI/PLL, no ST-LINK clock dependency



VCAP1 is the INTERNAL regulator output. Only its local capacitor connects here.
Keep every 100n beside its VDD pin; keep the VCAP loop short. DS10086 Fig.2 / Fig.18.



TEST: active low on PC13. 10k/100n = 1ms release RC.
Firmware must debounce and meet the 10ms local-fault deadline.
Reset and self-test are separate buttons.

SWD: pin1 VTref senses target voltage; do not feed a second 3.3V supply.
UART header is TX/RX/GND, not RS-232. NRST pulls low to reset.
BOOT0 low boots flash; fit recovery shunt only for bootloader recovery.

Shared rails and CAN backbone connect through the root sheet. SPI, clocks and reset stay within each zone.

+3V5 +3V3 +6V6_SERVO +6V6_SERVO GND GND CAN_A_H CAN_A_H CAN_A_L CAN_A_L CAN_B_H CAN_B_H CAN_B_L CAN_B_L

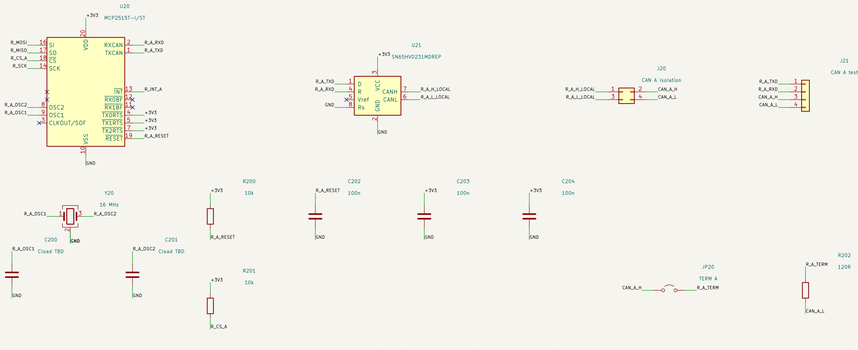
Engineering review: actuator/protection and mechanical sign-off pending.
PRELIMINARY — incomplete board: not for fabrication

Team 4	
Sheet: /Front zone/	
File: FrontLklcad_sch	
Title: Front zone — Integrated logic and interfaces	
Size: A1	Date: 2026-10-20
KiCad E.D.A. 10.0.4	Id: 2/5

Rev: 0.3

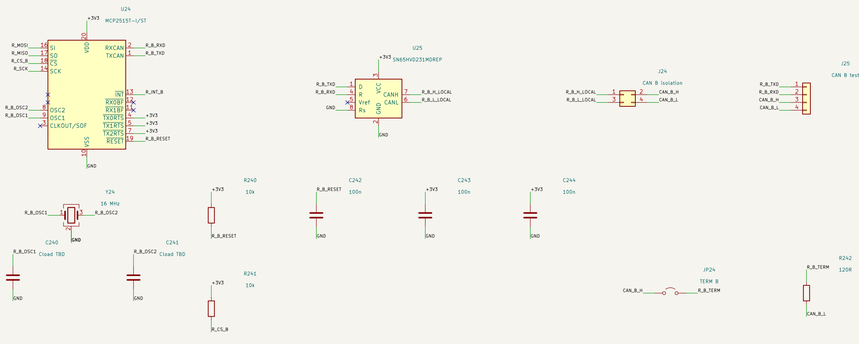
Rear zone — CAN and motor current sensing

CAN A: 16 MHz / 500 kbit/s / 3.3 V



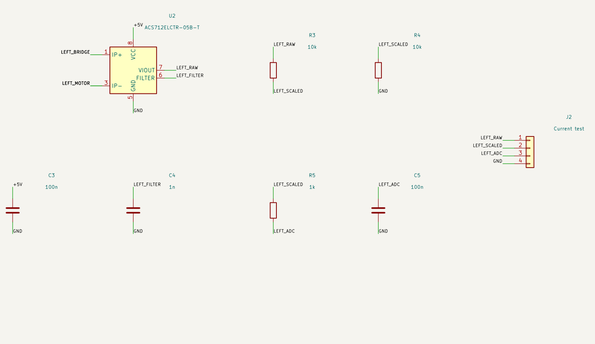
RS = GND: high-speed mode, Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

CAN B: 16 MHz / 500 kbit/s / 3.3 V

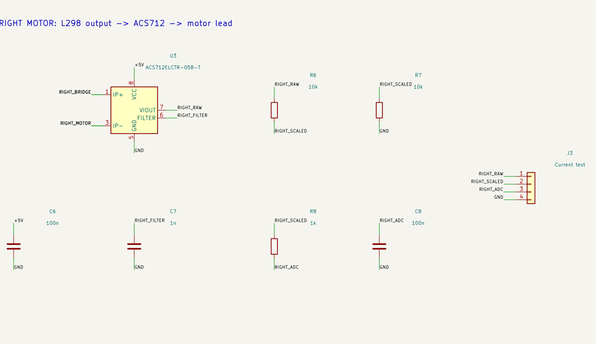


RS = GND: high-speed mode, Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

LEFT MOTOR: L298 output -> ACS712 -> motor lead



RIGHT MOTOR: L298 output -> ACS712 -> motor lead



Each motor: +/−5 A; VADC = 1.25 V + 0.0925 V/A * I.
Ideal 12-bit scale: 8.71 mA/count at VREF=3.3 V.
10k/10k divider keeps a 5 V sensor output at 2.5 V.
Filter source resistance: (10k || 10k) + 1k = 6k.
6k * 100n -> 265 Hz. Proposed ADC: 4 ksample/s/channel.
Stop motors before zero calibration: preserve signed readings.
Hall sensor has internal conductor, not an external shunt.
At 5 A and typical 1.2 mOhm, conductor loss = 30 mW.
Motor return lead connects to the OTHER L298 output, not GND.

J603

LAB3 rows 1–13

GND	1	2
GND	3	4
R_RX_A_MARKER	5	6
R_RX_B_MARKER	7	8
R_TX_A_MARKER	9	10
R_TX_B_MARKER	11	12
R_DEG_A_LED	13	14
R_DEG_B_LED	15	16
R_PWM_SET_MARKER	17	18
R_LEFT_ENABLE_PWM	19	20
R_DIR_A_MARKER	21	22
R_REAR_LEFT_LED	23	24
R_REAR_RIGHT_LED	25	26

LAB3 header follows handout p17 top-to-bottom, two GND pins first.
SRV/PWM_OUT probe actual actuator PWM; timing markers remain separate.
Physical header position / accessibility will be checked during PCB layout.

J604

Additional probes

GND	1	2
R_SELFTEST_BUTTON	3	4
R_ERROR_LED	5	6
R_TASK_PROBE	7	8
R_STEER_TASK_MARKER	9	10
R_PID_MARKER	11	12
R_RIGHT_ENABLE_PWM	13	14
R_OVERLOAD_MARKER	15	16
LEFT_ADC	17	18
RIGHT_ADC	19	20

J605

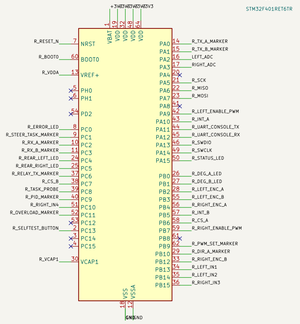
Reserved / encoder probes

GND	1	2
GND	3	4
R_LEFT_ENC_A	5	6
R_LEFT_ENC_B	7	8
R_RIGHT_ENC_A	9	10
R_RIGHT_ENC_B	11	12
R_RELAY_TX_MARKER	13	14

R630 10k BOOT0 pulldown
R631 10k PWM_EN pulldown
R632 10k NRST pulldown
R633 10k BOOT pulldown
R634 10k TEST pulldown
R635 10k RESET pulldown

Driver integration remains open pending overload/current-limit design.
PA9/PB7 are motor enables: low means coast during reset.
Motor/encoder connector pin order has not been inferred from cable colors.

STM32F401RET6 — independent HSI/PLL, no ST-LINK clock dependency



VCAP1 is the INTERNAL regulator output. Only its local capacitor connects here.
Keep every 100n beside its VDD pin; keep the VCAP loop short. DS10086 Fig.2 / Fig.18.

SWD: pin1 VTref senses target voltage; do not feed a second 3.3V supply.

UART header is TX/RX/GND, not RS-232. NRST pulls low to reset.
BOOT0 low boots flash; fit recovery shunt only for bootloader recovery.

J601

SWD (2.54mm)

+3V3	1	2
R_SWCLK	3	4
GND	5	6
R_SWDIO	7	8
R_RESET_N	9	10
GND	11	12

J602

3.3V TTL UART

R_UART_CONSOLE_TX	1	2
R_UART_CONSOLE_RX	3	4
GND	5	6

TEST: active low on PC13. 10k/100n = 1ms release RC.
Firmware must debounce and meet the 10ms local-fault deadline.
Reset and self-test are separate buttons.

Shared rails and CAN backbone connect through the root sheet. SPI, clocks and reset stay within each zone.

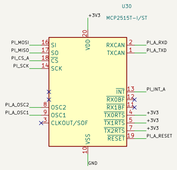
+3V3 +5V GND CAN_A_H CAN_A_L CAN_B_H CAN_B_L

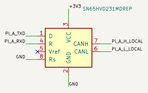
Engineering review: actuator/protection and mechanical sign-off pending.
PRELIMINARY — incomplete board; not for fabrication

Team 4	
Sheet: /Rear zone/	
File: Rear.kicad_sch	
Title: Rear zone — integrated logic and interfaces	
Size: A1	Date: 2026-10-20
KiCad E.D.A. 10.0.4	Rev: 0.3
	Id: 3/5

Cockpit zone — dual CAN

CAN A: 16 MHz / 500 kbit/s / 3.3 V







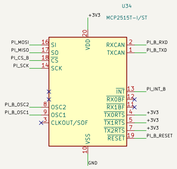


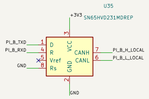


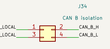


RS = GND: high-speed mode, Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

CAN B: 16 MHz / 500 kbit/s / 3.3 V







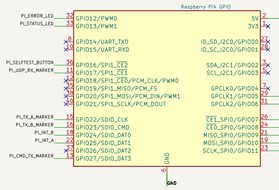




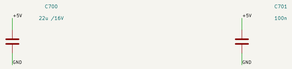


RS = GND: high-speed mode, Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

RASPBERRY PI 4 — 40-pin board socket



Pins 2/4 receive board 5.1V. PI 3.3V pins 1/17 are NOT joined to board 3.3V.
Do not attach independent USB-C power while board 5.1V is connected.
Confirm socket orientation with a 1:1 print and the actual PI before routing.



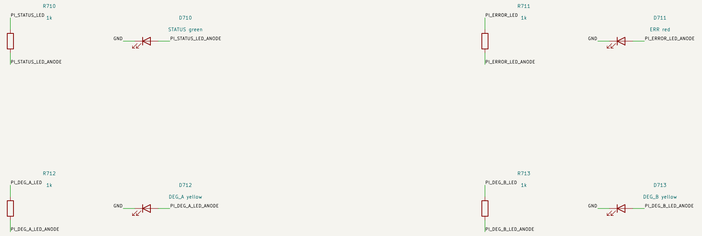
Local bypass is not a load-step proof. PI bulk capacitance / rail transient test remains open.



Rows 9-13 intentionally unused on PI (Lab3 p17).
Two GND pins per probe group. Active-high LED outputs.
Physical reachability is a layout check, not established by this schematic.







All PI GPIOs are 3.3V signals. Header pin number is NOT BCM GPIO number.
Independent board 3.3V supplies the six CAN interfaces.
A=CE0/IRQ25; B=CE1/IRQ24; confirm Linux labels against the installed overlay.

Shared rails and CAN backbone connect through the root sheet. SPI, clocks and reset stay within each zone.

+3V3 +3V3 +5V +5V GND GND CAN_A_H CAN_A_H CAN_A_L CAN_A_L CAN_B_H CAN_B_H CAN_B_L CAN_B_L

Engineering review; actuator/protection and mechanical sign-off pending.
PRELIMINARY — incomplete board; not for fabrication

Team 4

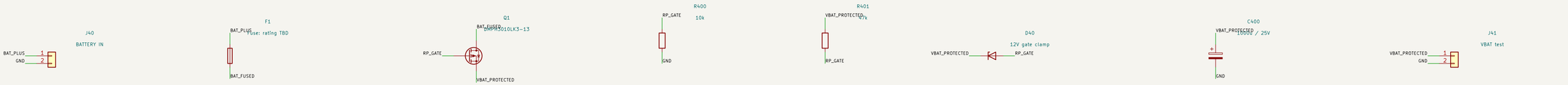
Sheet: /Cockpit zone/
File: Cockpit.kicad_sch

Title: Cockpit zone — integrated logic and interfaces

Size: A1	Date: 2026-10-20	**Rev: 0.3**
KiCad E.D.A. 10.0.4		Id: 4/5

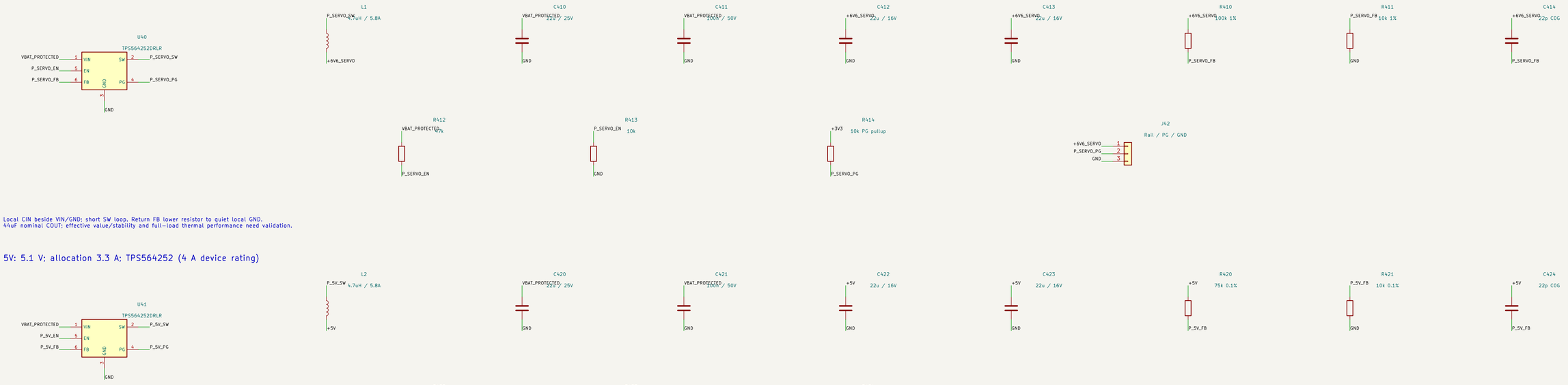
AA PACK INPUT — 10x NiMH recommended; exact cells/holder not selected

Design input envelope: 8–15 V at the protected node; verify real cells, contact resistance and sag.
Battery / motor rail is UNREGULATED. No lithium charger or lithium BMS is included.



Q1 loss at 10 A: 1 W using 10 mOhm at 25 C; hotter silicon has higher resistance.
F1/holder/cable limits, transient overvoltage and thermal layout are open release checks.
J40 generic footprint is a visible placeholder; DO NOT use a small signal header for battery current.

SERVO: 6.6 V; allocation 4 A; TPS564252 (4 A device rating)

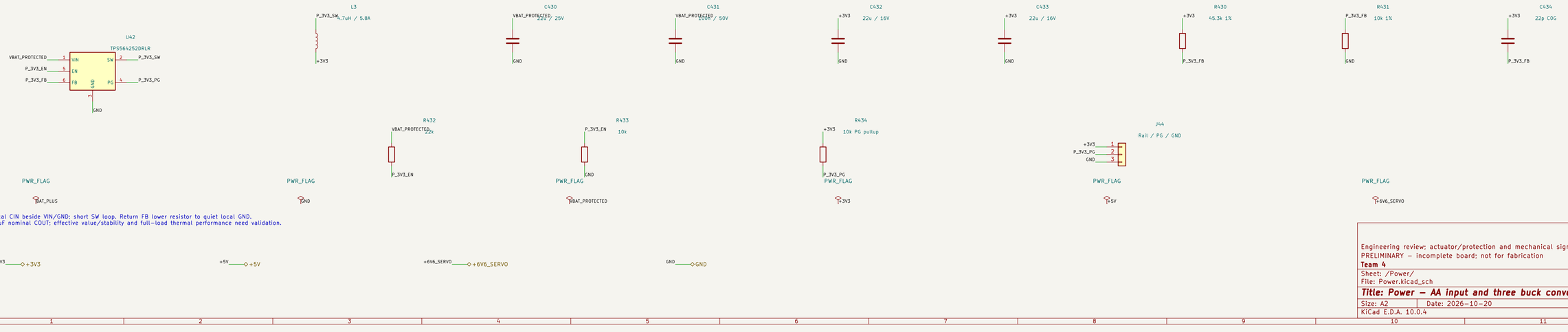


Local CIN beside VIN/GND; short SW loop. Return FB lower resistor to quiet local GND.
44uF nominal COUT; effective value/stability and full-load thermal performance need validation.

5V: 5.1 V; allocation 3.3 A; TPS564252 (4 A device rating)

Local CIN beside VIN/GND; short SW loop. Return FB lower resistor to quiet local GND.
44uF nominal COUT; effective value/stability and full-load thermal performance need validation.

3V3: 3.318 V; allocation 0.8 A; TPS564252 (4 A device rating)



Local CIN beside VIN/GND; short SW loop. Return FB lower resistor to quiet local GND.
44uF nominal COUT; effective value/stability and full-load thermal performance need validation.

+3V3 ◇ +3V3
+5V ◇ +5V
+6V6_SERVO ◇ +6V6_SERVO
GND ◇ GND

Engineering review: actuator/protection and mechanical sign-off pending.
PRELIMINARY — incomplete board; not for fabrication

Team 4

Sheet: /Power/
File: Power.kicad_sch

Title: Power — AA input and three buck converters

Size: A2	Date: 2026-10-20	**Rev: 0.3**
KiCad E.D.A. 10.0.4		Id: 5/5